

Social Science Inquiry II

Week 6: Linear models, part I

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Loading packages for this class

```
library(ggplot2)
```

- We often want to discuss the relationship between an *independent variable* and a *dependent variable*.
- One way to do this is to talk about a conditional mean; for example, if $X \in \{0, 1\}$, we may be interested in $E[Y|X = 0]$ and $E[Y|X = 1]$.
- (Which one is the *independent variable* and which is the *dependent variable*?)
- What if X takes on more than a few values?

Recall:

Angrist, Joshua D., and Alan B. Krueger. "Does compulsory school attendance affect schooling and earnings?" *The Quarterly Journal of Economics* 106.4 (1991): 979-1014.

```
dat <- read.csv('../data/angrist-krueger.csv', as.is = TRUE)

dat$year_of_birth_adj <- dat$year_of_birth +
  0.25 * (dat$quarter_of_birth-1)

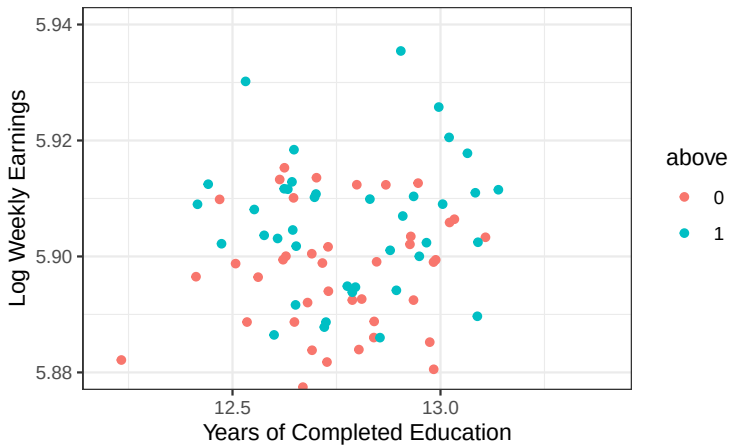
states_above_16 <- c(15, 23, 32, 35, 39, 40, 41, 42, 48, 49, 51, 53)
dat$states_above_16 <- 1 * (dat$place_of_birth %in% states_above_16)

dat_agg <- aggregate(x = dat[, c('log_weekly_wage', 'education')],
  by = list(`year_of_birth_adj` = dat$year_of_birth_adj,
    `quarter_of_birth` = dat$quarter_of_birth,
    `above` = as.factor(dat$states_above_16)),
  FUN = mean)
```

```
head(dat_agg)
```

```
##   year_of_birth_adj quarter_of_birth above log_weekly_wage education
## 1                30                1     0      5.882141    12.23273
## 2                31                1     0      5.898764    12.50745
## 3                32                1     0      5.888690    12.53485
## 4                33                1     0      5.892047    12.68044
## 5                34                1     0      5.888694    12.64883
## 6                35                1     0      5.877465    12.66922
```

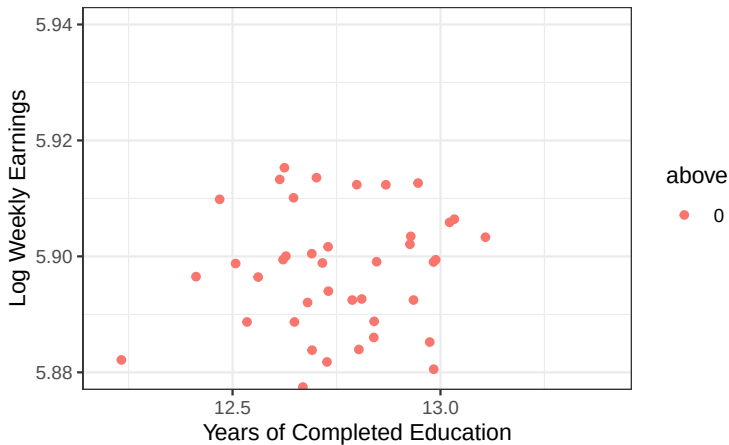
Angrist and Krueger, Earnings on Education by Age of Compulsory Schooling



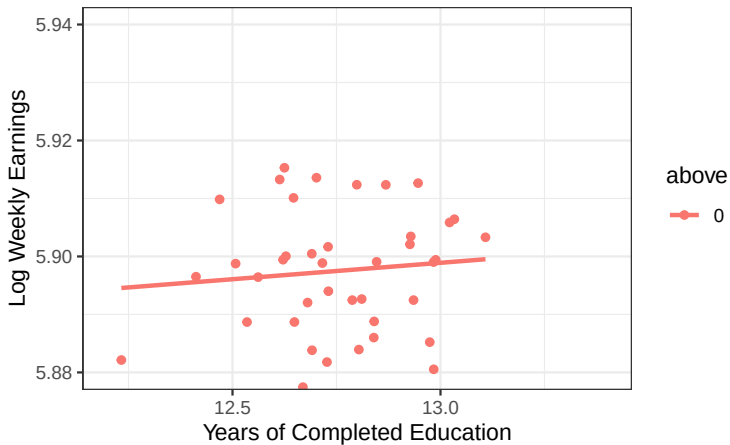
```
mean(dat_agg$log_weekly_wage[which(dat_agg$above == 0)])  
## [1] 5.897526  
  
mean(dat_agg$log_weekly_wage[which(dat_agg$above == 1)])  
## [1] 5.906715
```


- Suppose we want to draw a line through these points.
- What is the best way to pick the line?

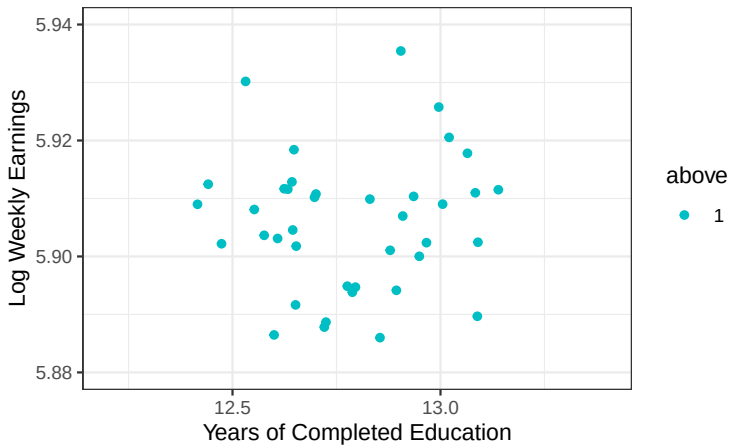
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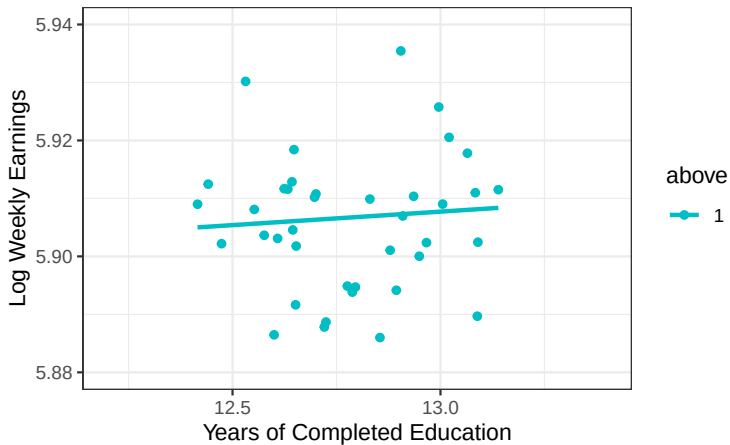
Angrist and Krueger, Earnings on Education by Age of Compulsory Schooling



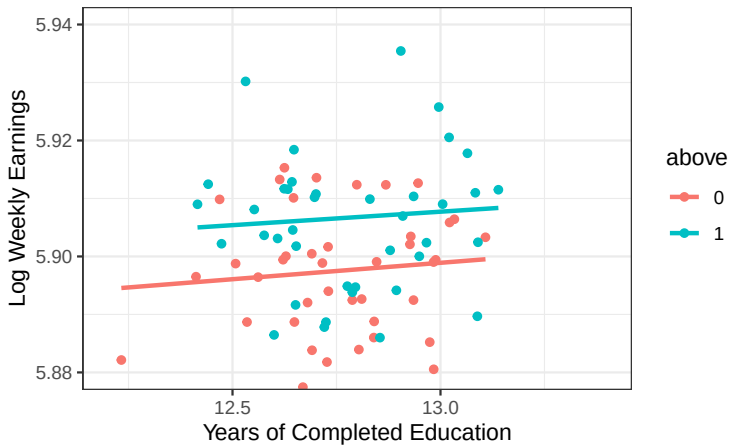
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- We would like to describe a conditional relationship in the data

$$E[Y|X = x] = h(x)$$

where one simple version is

$$g(x) = \beta_0 + \beta_1 x$$

- In other words,

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

where

$$E[\epsilon_i|X_i] = 0$$

and

$$\text{Var}[\epsilon_i|X_i] = \sigma^2$$

- In practice, we describe Y_i as a function of X_i in the data we *observe*.
- We refer to this as “regressing Y on X .”

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

- We can then define *residuals*

$$\hat{\epsilon}_i = Y_i - \hat{Y}_i = Y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 X_i \right).$$

- We calculate estimates of $\hat{\beta}_0$ and $\hat{\beta}_1$ as the values that minimize the residual sums of squares

$$\text{RSS} = \sum_{i=1}^n \hat{\epsilon}_i^2$$

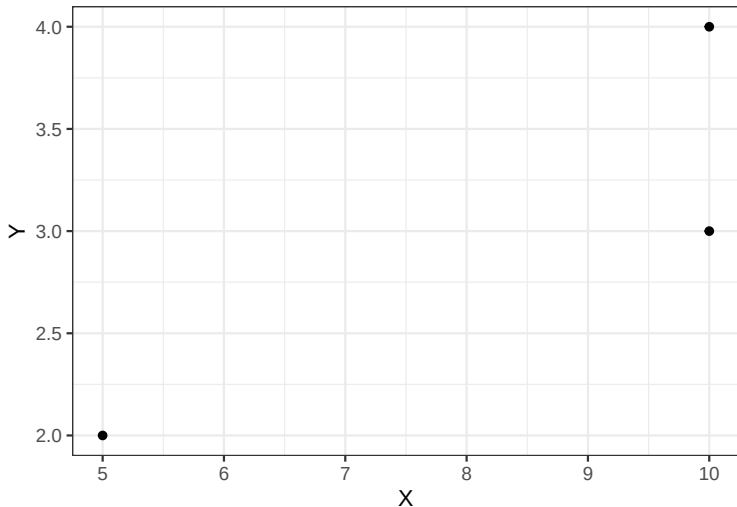
Suppose we had the following data points:

```
toy_dat <- data.frame(Y = c(2, 3, 4),  
                      X = c(5, 10, 10))
```

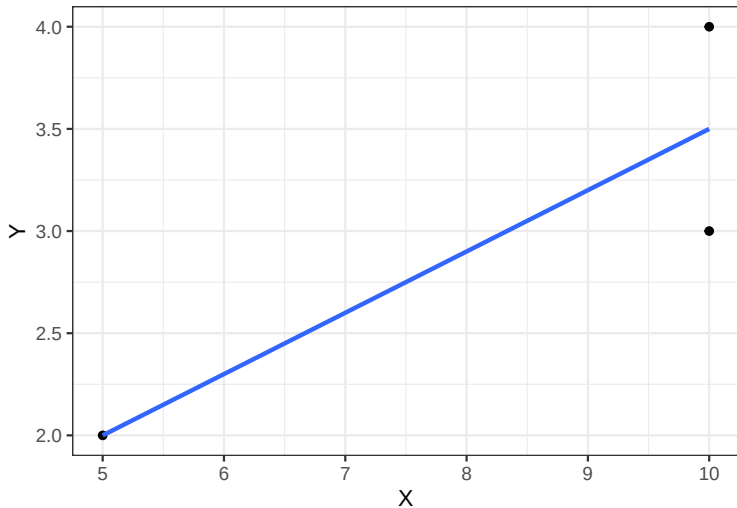
```
toy_dat
```

```
##      Y  X  
## 1 2  5  
## 2 3 10  
## 3 4 10
```

What values for $\hat{\beta}_0$ and $\hat{\beta}_1$ minimize the residual sum of squares?



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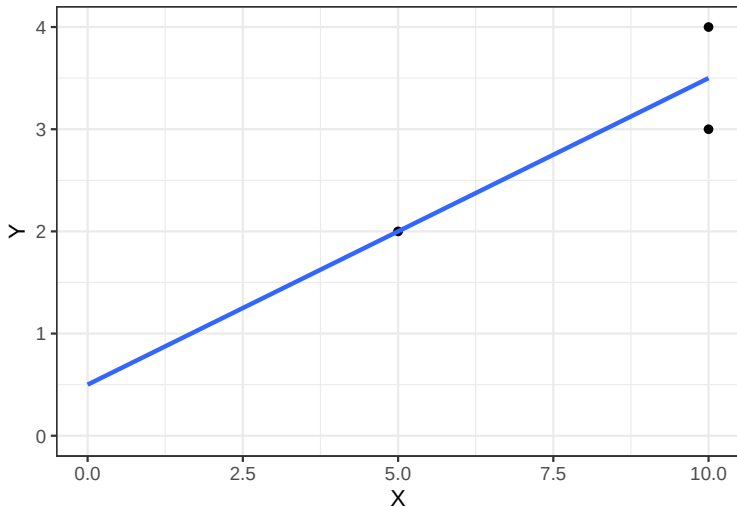


We can also think about $\hat{\beta}_0$ and $\hat{\beta}_1$ as the *y-intercept*, i.e., where the line crosses the y-axis, and the *slope*, respectively.

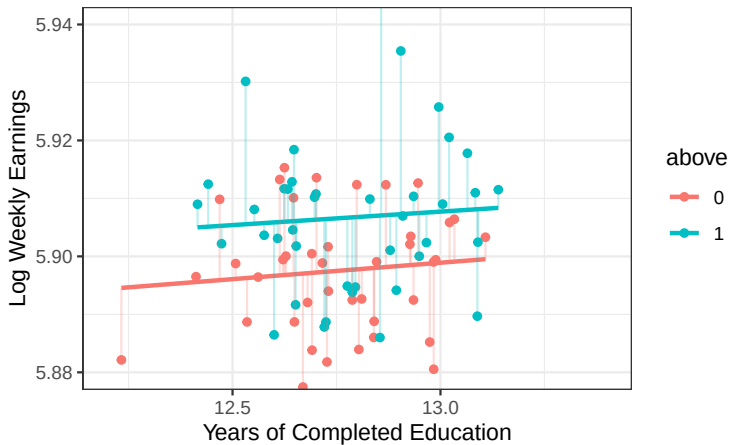
```
lm(Y~X, data = toy_dat)

##
## Call:
## lm(formula = Y ~ X, data = toy_dat)
##
## Coefficients:
## (Intercept)          X
##          0.5          0.3
```

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- Why do we minimize the sum of *squared* distances? (rather than...absolute distances? Or cubed distances?)

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- According to IMS:
 - *It is the most commonly used method.*
 - *Computing the least squares line is widely supported in statistical software.*
 - *In many applications, a residual twice as large as another residual is more than twice as bad. For example, being off by 4 is usually more than twice as bad as being off by 2. Squaring the residuals accounts for this discrepancy.*
 - *The analyses which link the model to inference about a population are most straightforward when the line is fit through least squares. (What does this mean??)*

- Other potential reasons...
 - Squared distances will always be positive (so will absolute distances)
 - But absolute distances don't provide a unique solution to the minimization problem, squared distances do
 - It's easier to take the derivative of the square, rather than absolute.
 - **Minimizing RSS gives a linear approximation to the conditional expectation function.** (Why is it only an approximation? When is it not approximate?)

- Why do we take the squared distance in terms of Y —instead of in terms of X ?
- What if we regressed $X \sim Y$ instead of $Y \sim X$?

Returning to Butler and Broockman...

Butler, Daniel M., & Broockman, David E. (2011). Do politicians racially discriminate against constituents?
A field experiment on state legislators.

Data is available at the Yale ISPS data archive:
isps.yale.edu/research/data

Loading the data

```
df <- read.csv('../data/butler-broockman.csv', as.is = TRUE)
head(df)
```

##	leg_party	leg_republican	leg_black	leg_latino	reply_atall	treat_deshawn
## 1	R	1	0	0	1	0
## 2	D	0	0	0	1	1
## 3	R	1	0	0	0	1
## 4	R	1	0	0	0	1
## 5	D	0	0	0	0	0
## 6	D	0	0	0	1	1

##	treat_demprimary	treat_repprimary	treat_noprimary	treat_group	treat_jake
## 1	1	0	0	5	1
## 2	0	1	0	2	0
## 3	0	0	1	6	0
## 4	0	0	1	6	0
## 5	0	0	1	0	1
## 6	0	1	0	2	0

##	leg_notwhite	leg_white	leg_notblackotherminority	treat_primary
## 1	0	1	0	1
## 2	0	1	0	1
## 3	0	1	0	0
## 4	0	1	0	0
## 5	0	1	0	0
## 6	0	1	0	1

Recall that treatment is 1 if the sender was DeShawn Jackson, and 0 if Jake Mueller.

```
table(df$treat_deshawn)
```

```
##
```

```
##      0      1
```

```
## 2431 2428
```

The primary outcome is whether legislators replied at all.

```
table(df$reply_atall)
```

```
##
```

```
##      0      1
```

```
## 2112 2747
```

We're going to manipulate our data so it takes the format $Y \sim W$.

```
df$W <- df$treat_deshawn  
df$Y <- df$reply_atall
```

To get the difference-in-means estimate of the ATE,

```
Y1 <- df$Y[which(df$W == 1)]  
Y0 <- df$Y[which(df$W == 0)]
```

```
(dm_hat <- mean(Y1) - mean(Y0))  
## [1] -0.01782424
```

Legislators were 1.7 percentage points less likely to reply to an email if the sender was identified as DeShawn Jackson as compared to Jake Mueller.

What is the relationship with the conditional means?

```
lm(Y~W, data = df)

##
## Call:
## lm(formula = Y ~ W, data = df)
##
## Coefficients:
## (Intercept)          W
##      0.57425      -0.01782
```


- How do we interpret the coefficient on W ?
- The intercept?

Credit to Andy Eggers...

- The *fitted* regression can be written as

$$\hat{Y}_i = .574 - .018W_i$$

- We can express the conditional means as:

$$\hat{Y} = \begin{cases} .574 & W_i = 0 \\ .574 - .018 & W_i = 1 \end{cases}$$

Credit to Andy Eggers...

- With an intercept, the Democratic mean is the intercept and the Republican contrast is the slope:

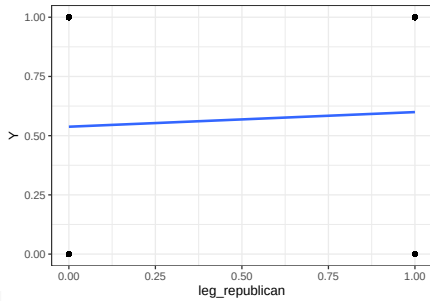
$$\hat{Y}_i = 0.538 + 0.062W_i$$

- Without an intercept, each coefficient is a group mean:

$$\hat{\mu}_D = 0.538, \quad \hat{\mu}_R = 0.6$$

- These two parameterizations imply the same fitted values.

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Minimizing the sum of squared distances ...

```
table(df$Y[which(df$leg_republican == 0)])  
  
##  
##      0      1  
## 1243 1446  
  
mean(df$Y[which(df$leg_republican == 0)])  
  
## [1] 0.5377464
```

```
table(df$Y[which(df$leg_republican == 1)])  
  
##  
##      0      1  
##  869 1301  
  
mean(df$Y[which(df$leg_republican == 1)])  
  
## [1] 0.5995392
```

...reproduces exactly the conditional means with a binary independent variable.

Extracting components from an lm object

```
lm1 <- lm(Y ~ W, data = df)
names(lm1)[1:8]

## [1] "coefficients" "residuals"      "effects"        "rank"
## [5] "fitted.values" "assign"          "qr"             "df.residual"

coef(summary(lm1))

##              Estimate Std. Error   t value Pr(>|t|)
## (Intercept)  0.57424928 0.01005440  57.114233 0.0000000
## W            -0.01782424 0.01422346  -1.253158 0.2102086
```

References I

Butler, D. M. and Broockman, D. E. (2011). Do politicians racially discriminate against constituents? a field experiment on state legislators. *American Journal of Political Science*, 55(3):463–477.